Introduction - Homework 1

- Backwards Heat Equation
- Abel Problem
- Deblurring Problem
- Quick overview of math along the way
- Likely will not be able to cover all
  - Will go through most part quite quickly
  - Feel free to ask more detailed questions during or after
- Feedback received on report
  - Use *FFT* for better efficiency (backwards heat problem)
  - Use *augmented least-squares* for improved stability (Tikhonov)

Backwards Heat Equation

Forward model:

$$Kx(t) = \int_0^\pi x(s)k(t,s)\,\mathrm{d}s \qquad\qquad k(t,s) = \frac{2}{\pi}\sum_{n=1}^{\infty} e^{-Tn^2}\sin(nt)\sin(ns)$$

Notice higher-order modes $n \gg 1$ are particularly attenuated

$\Longrightarrow$ inverse problem ill-posed (sensitivity).

Backwards Heat Problem — Singular System

$$Kx(t) = \int_0^\pi x(s)k(t,s)\,\mathrm{d}s \qquad k(t,s) = \frac{2}{\pi}\sum_{n=1}^{\infty} e^{-Tn^2}\sin(nt)\sin(ns)$$

Is given system the singular system?

$$x_j(t) = \sqrt{2/\pi}\sin(jt), \qquad y_j(s) = \sqrt{2/\pi}\sin(js), \qquad \mu_j = e^{-Tj^2}$$

We find this satisfies requirements for singular system:

- x_j, y_j span $L^2(0,\pi)$ (fourier basis)
- $Kx_j(t) = \int_0^\pi \sqrt{\frac{2}{\pi}}\sin(js)\left(\frac{2}{\pi}\sum_{n=1}^{\infty} e^{-Tn^2}\sin(nt)\sin(ns)\right)\mathrm{d}s$

$$= \left(\tfrac{2}{\pi}\right)^{\frac{3}{2}}\left(\sum_{n=1}^{\infty} e^{-Tn^2}\sin(nt)\int_0^\pi \sin(js)\sin(ns)\,\mathrm{d}s\right) = \left(\tfrac{2}{\pi}\right)^{\frac{3}{2}}\left(\sum_{n=1}^{\infty} e^{-Tn^2}\sin(nt)\tfrac{\pi}{2}\delta_{jn}\right)$$

$$= e^{-Tn^2}\sqrt{\tfrac{2}{\pi}}\sin(jt) = \mu_j y_j(t)$$

  - Tricks: orthogonality $\sin(js)\sin(ns) = \delta_{jn}$; Gaussian decay $\Rightarrow$ uniform convergence
- Similarly $K^* y_j = \mu_j x_j$ upon finding $K = K^*$ due to symmetric, real kernel: $k^*(s,t) = k(t,s)$
- $\mu_j > 0$ and $\mu_j \to 0$ as $j \to \infty$

$\Rightarrow$ *singular system ✓*

Backwards Heat Problem — Discretisation

We discretise operator K into $A \in \mathbb{R}^{n \times n}$ with $n = 128$ using either **FFT** $\sim \mathcal{O}(n \log n)$, or **sampling of kernel** at grid points $\sim \mathcal{O}(n^2)$.

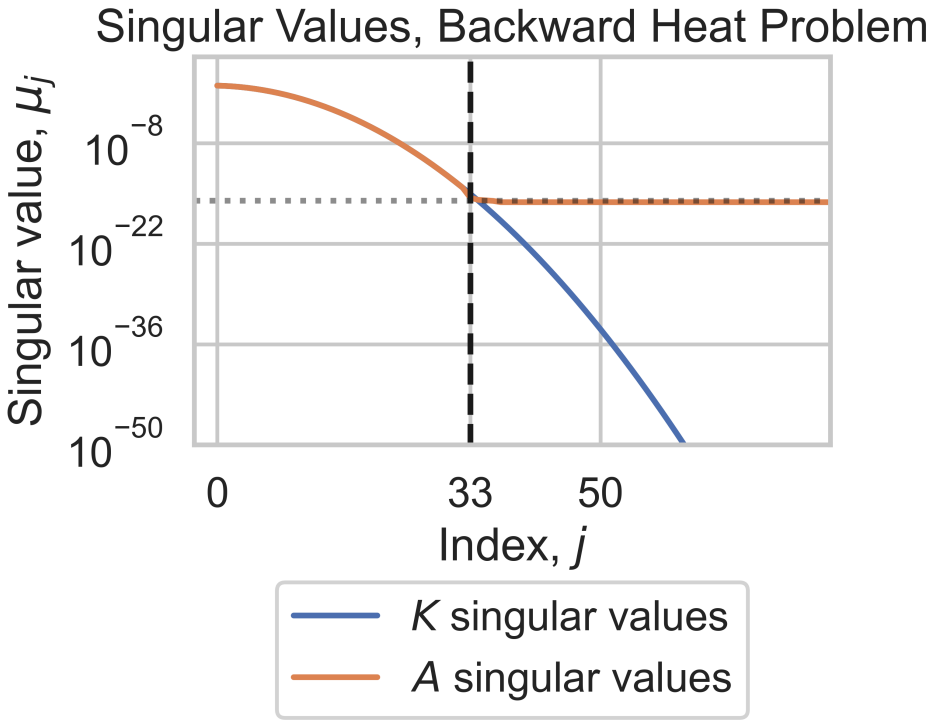
For this assignment we did the latter, constructing $A = U \Sigma V^{\mathbb{T}}$

$$U_{ij}(t) = \sqrt{\frac{2}{n+1}} \sin\left(j \cdot t_j\right) \qquad V_{ij}(s) = \sqrt{\frac{2}{n+1}} \sin\left(j \cdot s_j\right) \qquad \Sigma_{jj} = e^{-Tj^2}$$

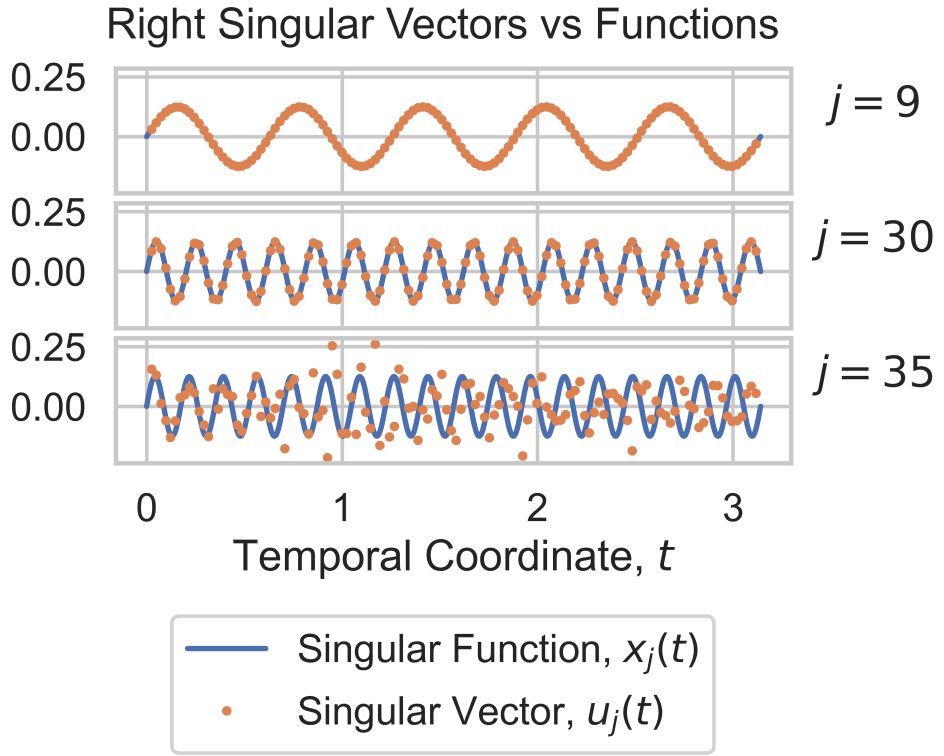
Notes:

- Possible as we know singular system analytically.
  - Generally requires quadrature.
- Normalisation in pre-factor
- For other assignments, the FFT approach was used instead
  - Matrix-free implementation used in HW3.

Backwards Heat Problem — Accuracy of Discretisation



Note cut-off at machine precision, $\sim 10^{-16}$, corresponding to $j \approx 33$

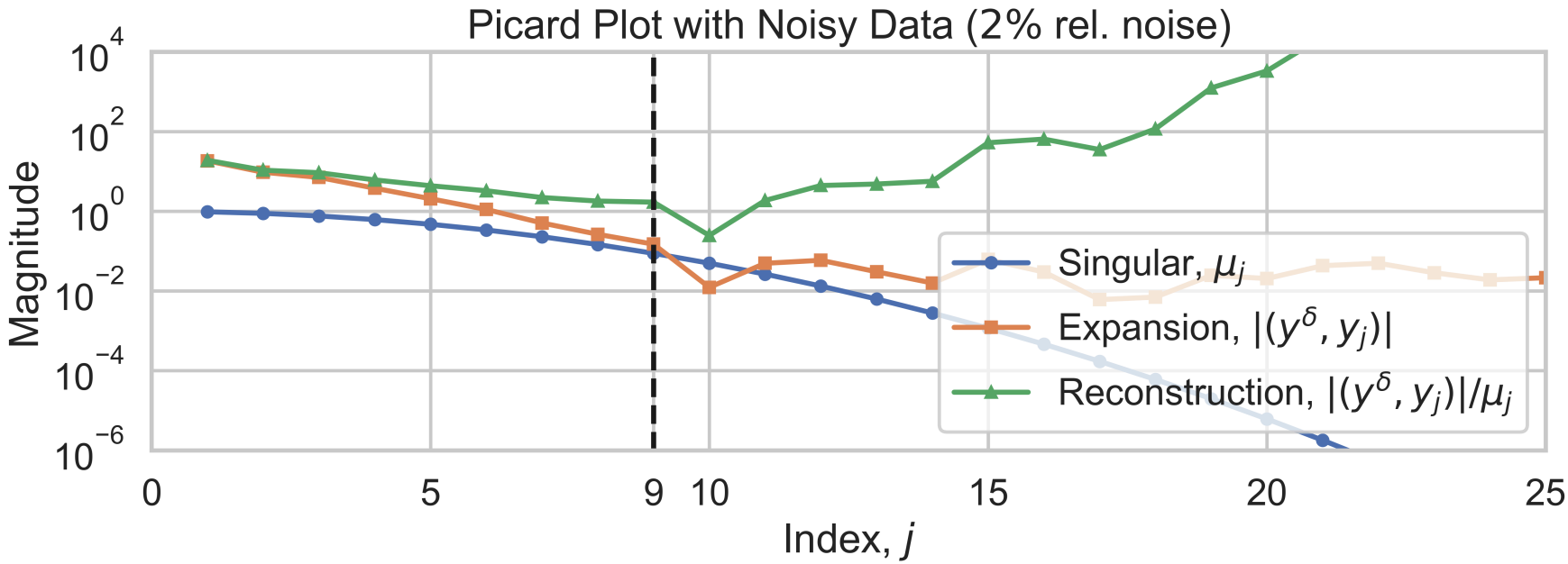


Last mode is past $j = 33$ cut-off, so noise dominates decomposition

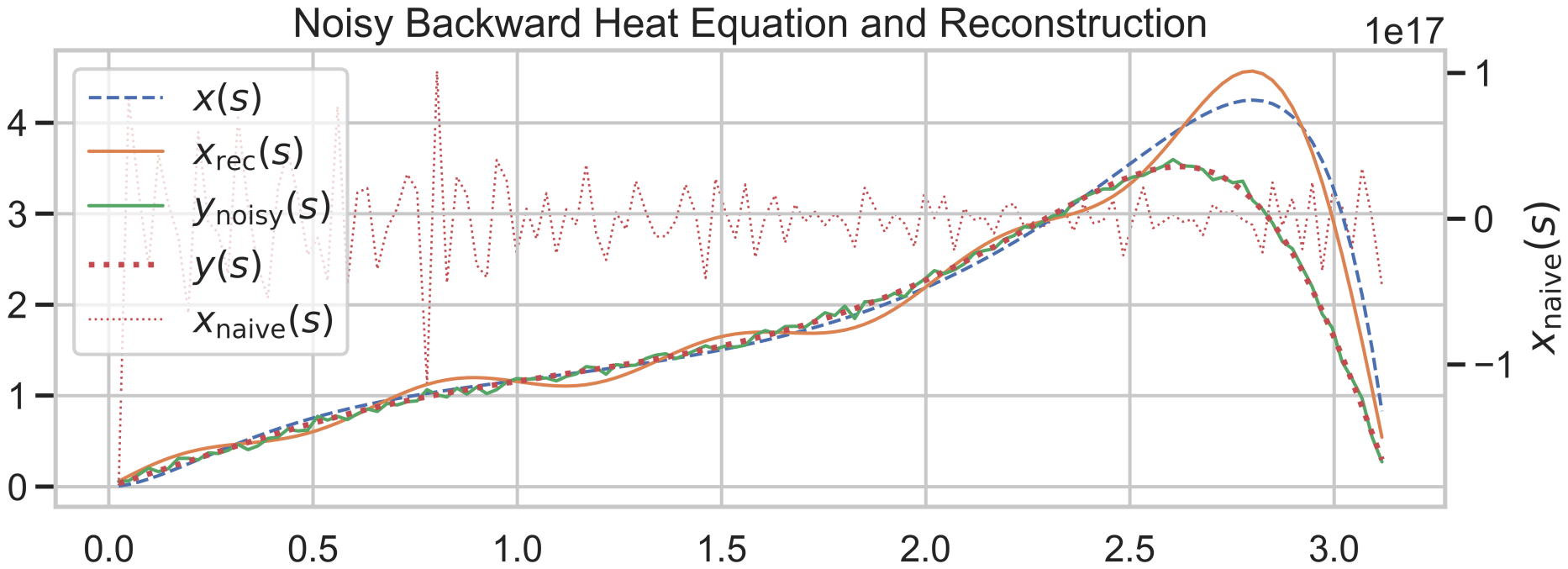
Backwards Heat Problem — Truncated SVD

Given function $x(s) = s\sin(s)\exp\left(\left(s - \frac{\pi}{2}\right)^2\right)$ invert noisy measurements y^δ where noise level $\delta = 2\%$

Truncated SVD (TSVD) to regularise inversion, only first $j^\star$ modes are used.
$j^\star = 9$ found using **Picard Plot** and **Discrete Picard Condition**: *expansion coefficients must decay faster than singular values.*



Backwards Heat Problem — Reconstruction Result



- **Note**: x_{naive} is read on right axis. It represents naive inversion using A^{-1}

Abel Problem

We consider the Abel Problem defined in $L^2(0,1)$:

$$Kx(t) = \int_0^t \frac{x(s)}{\sqrt{t-s}}\,\mathrm{d}s$$

Proof of compactness skipped here, but outlined as:

- Singularity handled by offset kernel K_ε
- Proof then becomes $K_\varepsilon \to K$ as $\varepsilon \to 0$. Use difference operator, Young's convolution inequality

Noisy measurements are produced analytically using $y^\delta = y + n$ where n corresponds to 5% Gaussian noise.

We attempt to reconstruct using both TSVD and Tikhonov regularisation:

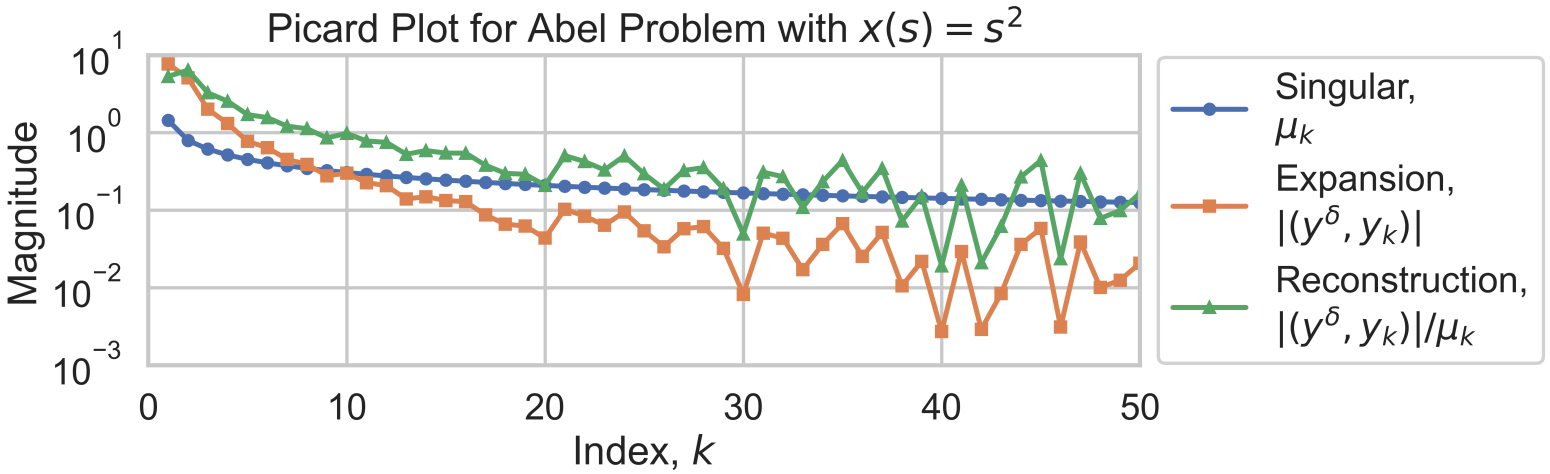
$$x_\alpha = (K^*K + \alpha I)^{-1} K^* y^\delta = (A^T A + \alpha I)^{-1} A^T y^\delta$$

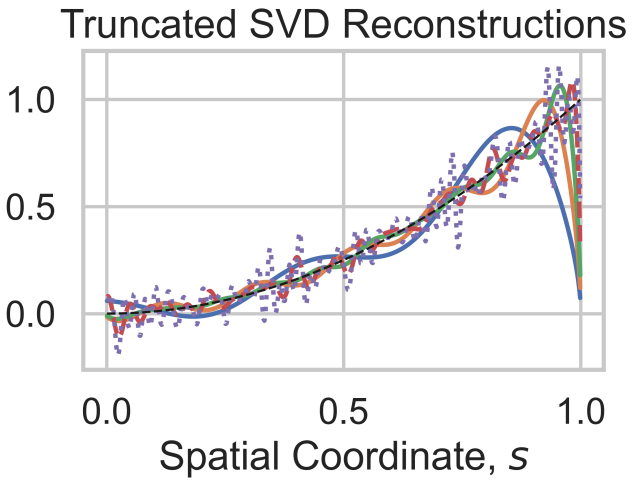
Note: improved numerical stability when using **augmented least-squares** formulation, which may be solved using `np.linalg.lstsq`:

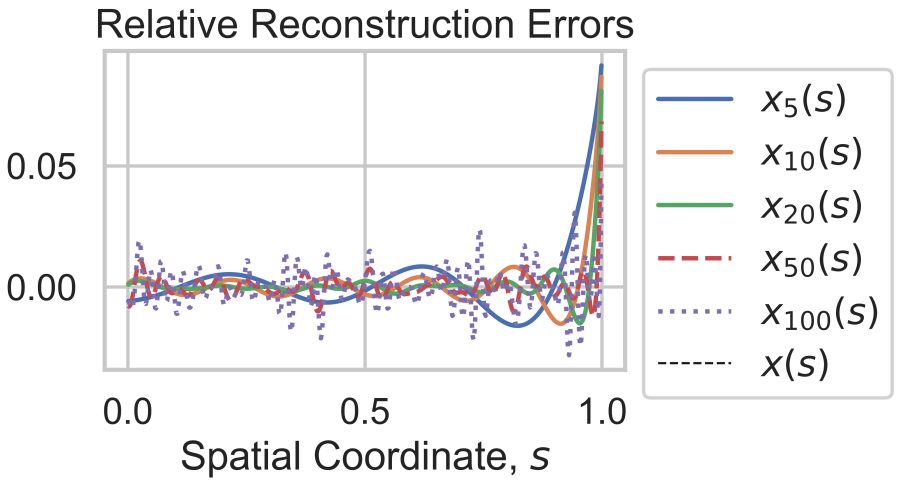
$$\tilde{A} = \left[A; \sqrt{\alpha} I\right], \qquad \tilde{y} = \left[y^\delta; 0\right]$$

This arises from the Tikhonov variational formulation, $\min_x \left\| Ax - y^\delta \right\|_2^2 + \alpha \|x\|_2^2$

Abel Problem — TSVD Reconstruction



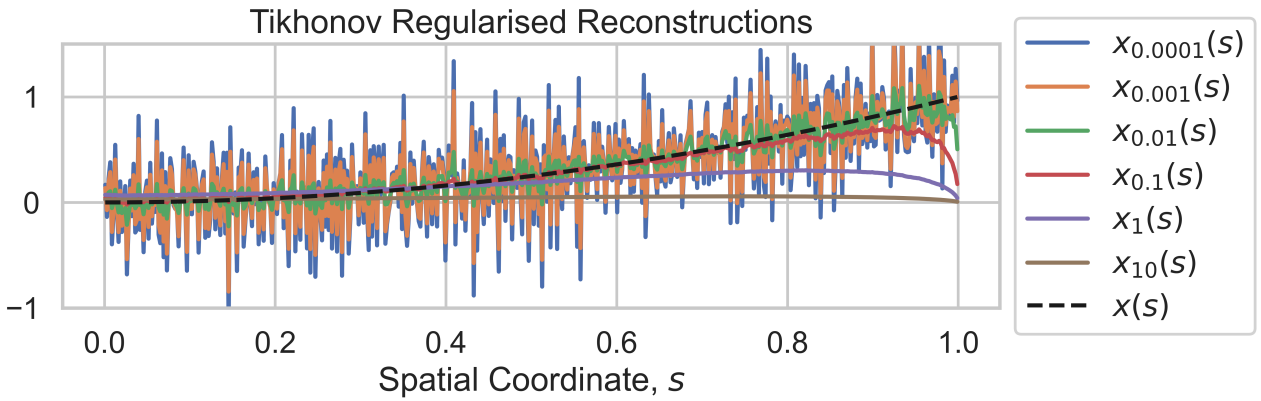


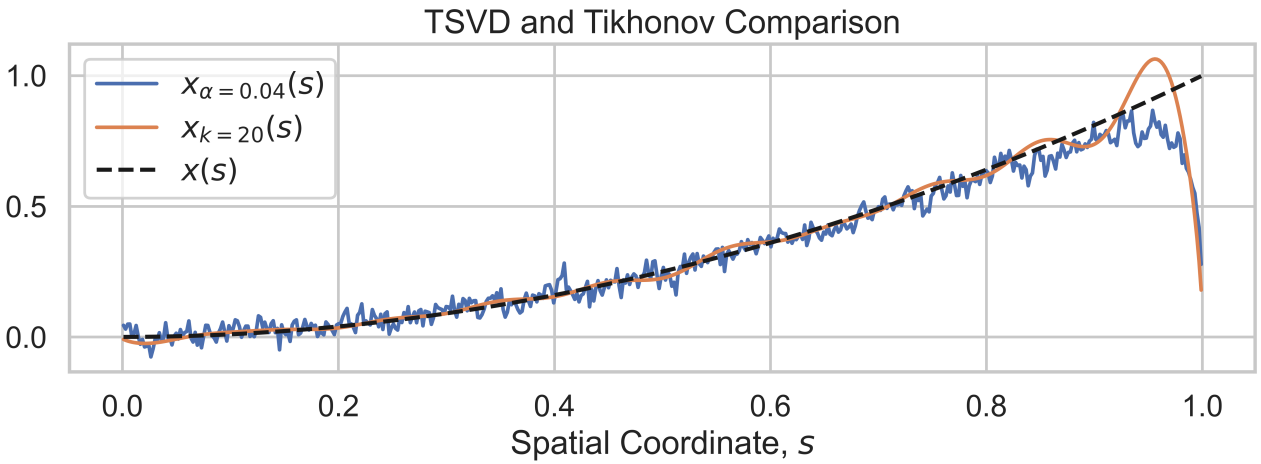


Notes:

- Picard $\Rightarrow$ $j^\star \approx 20$
- Strong bias as $s \to 1$

Abel Problem — Tikhonov Reconstruction





- Both biased, Tikhonov less so
- Very smooth true solution, so truncation works well
- As always, **variance-bias trade-off**

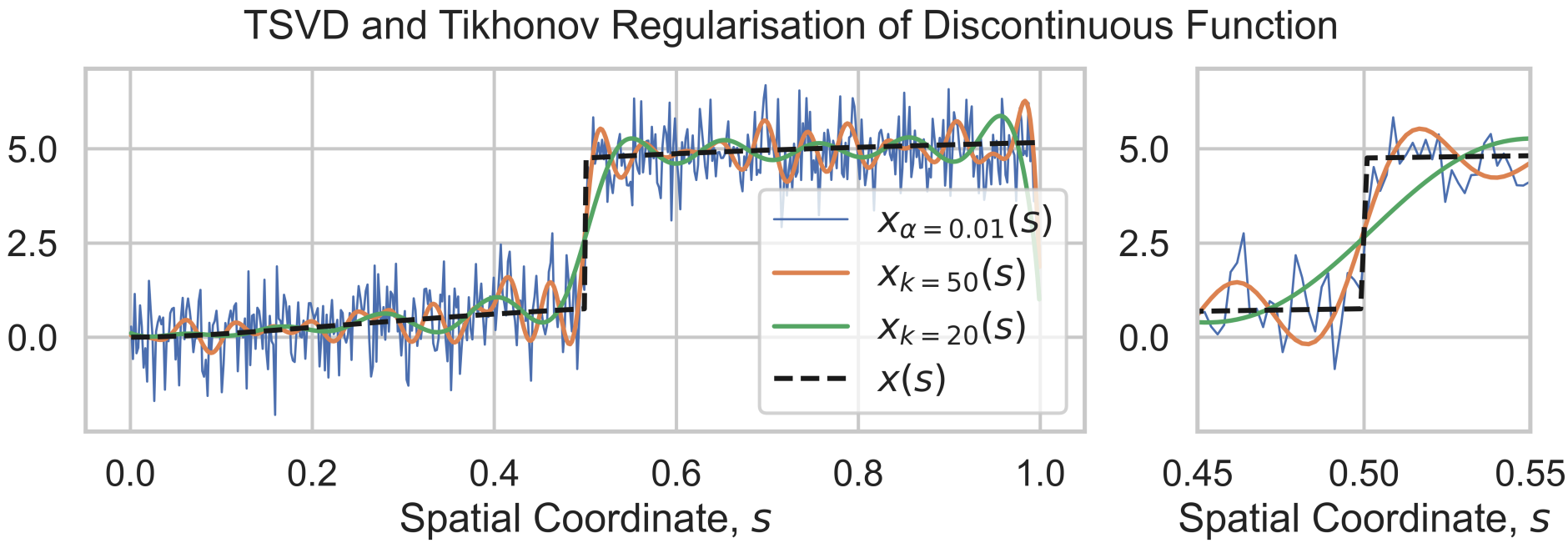
Abel Problem — Proving Bias at $s = 1$

It should be possible to probe bias at $s = 1$ analytically.

- First attempt in discrete case failed
- Continuous case easier, outline:
  - Consider $\left(K^*Kx_j\right)(s=1) = \mu_j^2 x_j(s=1)$
  - Find adjoint operator $K^*y(s) = \int_s^1 \frac{y(t)}{\sqrt{t-s}}\,\mathrm{d}t$ using inner-products and Fubini's theorem
  - Thus we find: $\left(K^*Kx_j\right)(s=1) = \mu_j^2 x_j(s=1) = \underbrace{\int_{s=1}^{1}}_{=0} \int_0^t \frac{1}{\sqrt{t-\tau}} y(\tau)\,\mathrm{d}\tau = 0$
  - Since $\mu_j \neq 0$, we must have $x_j(s=1) = 0$ for all j
- Argue that discrete case must tend to this as $n \to \infty$

Abel Problem — Discontinuous Function

We also investigate a discontinuous function:



- High frequency components required to capture discontinuity
- This means less effective noise reduction

Deblurring Problem

We are given a regularisation operator $\tilde{R}_\alpha$ such that

$$x_\alpha = \tilde{R}_\alpha y = \sum_j \frac{\mu_j^2}{\mu_j^3 + \alpha}(y, y_j)x_j$$

For this to be a regulariser, we require $\tilde{R}_\alpha K x = x$ as $\alpha \to 0$. Outline of proof:

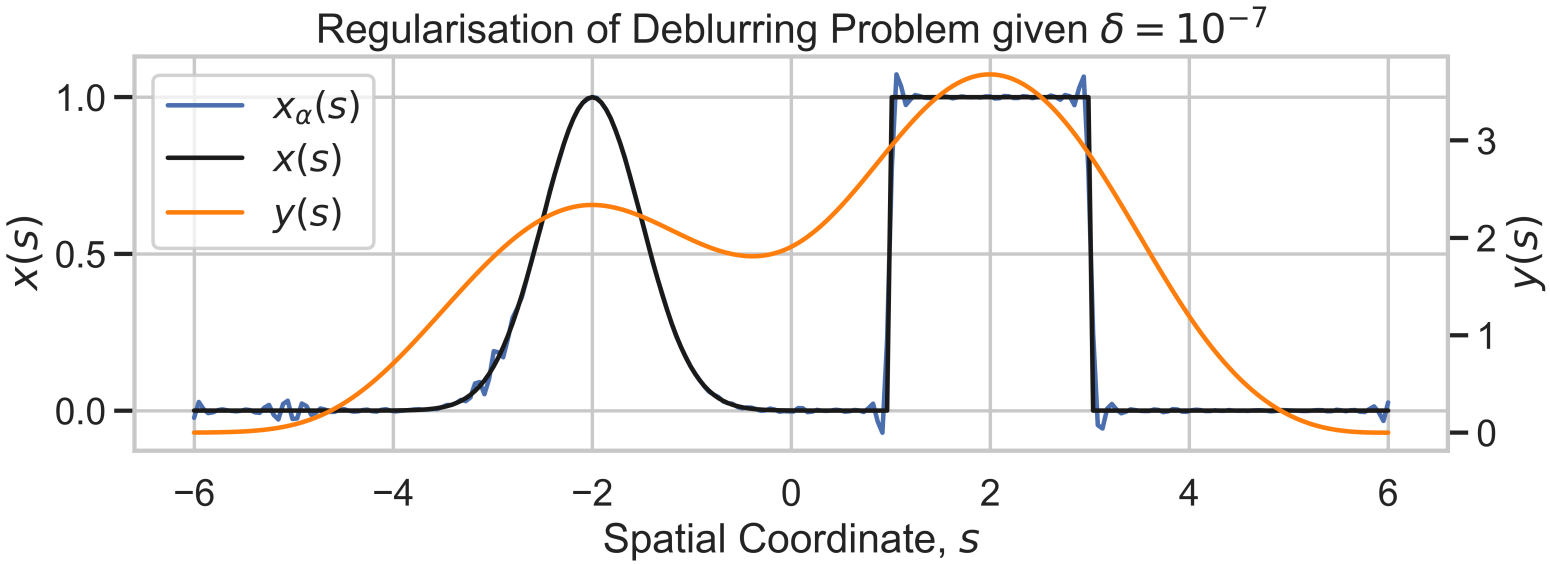
- Decompose x in singular system: $x = \sum_j (x, x_j)x_j, \qquad y = Kx = \sum_j \mu_j (x, x_j)y_j$ (complete basis, injectivity)
- Find $(y, y_j) = \mu_j (x, x_j)$ by linearity of inner product (sum-integral swap) and orthonormality of basis
- Consider norm of difference, $\left\|\tilde{R}_\alpha K x - x\right\|^2 = \left\|\sum_j \left[\left(\frac{\mu_j^3}{\mu_j^3+\alpha}\right) - 1\right](x, x_j)x_j\right\|^2$
- Apply Plancherel theorem: $\sum_{k=1}^{\infty} \left|(x, x_k)_X\right|^2 = \|x\|_X^2$
- Take limit $\alpha \to 0^+$
- Use *Lebesque's Dominated Convergence Theorem* to interchange limit and sum
- Thus we find that the term factor $\frac{\mu_j^3}{\mu_j^3+\alpha} - 1 \to 0$ as $\alpha \to 0, \forall j$
- Hence $\left\|\tilde{R}_\alpha K x - x\right\| \to 0$ as $\alpha \to 0$ ■

Deblurring Problem — Convergence

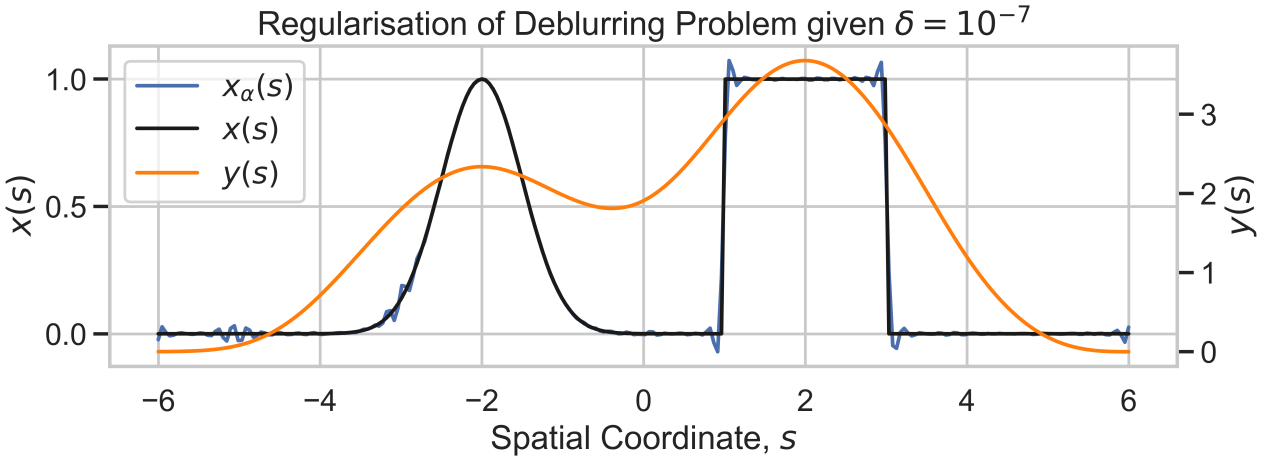
We test convergence of regulariser using blurring problem

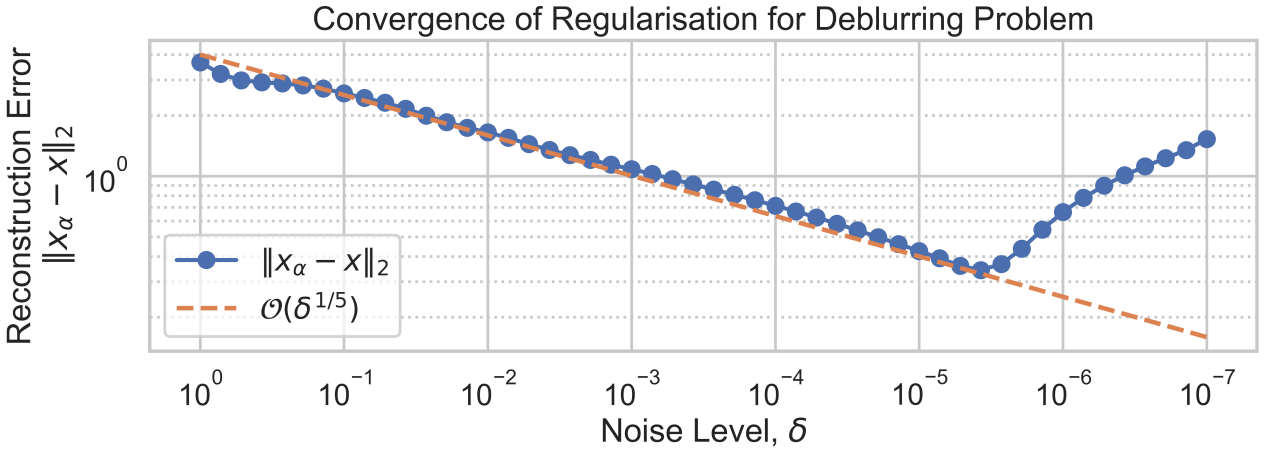
$$Kx(t) = \int_{-6}^{6} k(t-s)x(s)\,\mathrm{d}s \qquad \text{where} \qquad k(z) = \begin{cases} 1+\cos\left(\pi\frac{z}{3}\right) & \text{if} \quad |z| < 3 \\ 0 & \text{if} \quad |z| \geq 3 \end{cases}$$

In report, we picked $\alpha(\delta) = \delta^3$, though it should be $\alpha(\delta) = \delta^2$, which we investigate the convergence of here for $\delta = 10^{-7}$, which determines the noise level.



Deblurring Problem — Convergence Results





- Good convergence observed
- Changing from $\alpha = \delta^3$ to $\alpha = \delta^2$ does not change rate of convergence or results
- Diverges after $\delta \approx 6 \times 10^{-6}$ due to numerical stability